

# Technical University of Denmark

**Written examination:** 23rd May 2023, 9 AM — 1 PM.

**Course name:** Introduction to Machine Learning and Data Mining.

**Course number:** 02450.

**Aids allowed:** All aids permitted.

**Exam duration:** 4 hours.

**Weighting:** The individual questions are weighted equally.

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The exam is multiple choice. All questions have four possible answers marked by the letters A, B, C, and D as well as the answer “Don’t know” marked by the letter E. Correct answer gives 3 points, wrong answer gives  $-1$  point, and “Don’t know” (E) gives 0 points.

**This exam only allows for electronic hand-in.**

You hand in your answers at <https://eksamen.dtu.dk/>. To hand in your answers, write them in the file `answers.txt` (this file is available from the same place you downloaded this file). When you are done, upload the `answers.txt` file (and nothing else). Double-check that you uploaded the correct version of the file from your computer.

**Do not change the format of `answers.txt`**

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| No.   | Attribute description             | Abbrev. |
|-------|-----------------------------------|---------|
| $x_1$ | Peak-to-average power ratio       | P2AR    |
| $x_2$ | Periodicity                       | PER     |
| $x_3$ | Roughness                         | RO      |
| $x_4$ | Fluctuation                       | FLU     |
| $x_5$ | Sharpness                         | SHP     |
| $x_6$ | Loudness                          | LN      |
| $y$   | Sound class                       |         |
| $y_r$ | Percieved annoyance measure (PAM) |         |

Table 1: The dataset contains 564 audio recordings from two classes, Machine and Natural sounds. Machine sounds are produced by man-made machinery such as cars or chainsaws. Natural sounds result from natural phenomena such as rain or dogs barking. Additionally, a perceived annoyance measure (PAM),  $y_r$ , has been observed, quantifying the annoyance perceived by a human listener. Each recording has been preprocessed to extract 6 attributes representing the recording. The attributes are described in the table above. The classification task concerns the prediction of the sound type,  $y$  (Machine or Natural), while the regression task concerns the prediction of the percieved annoyance level,  $y_r \in \mathbb{R}$ .

**Question 1.** The main dataset used in this exam is the Sound Classification dataset<sup>1</sup> described in Table 1. Figure 1 shows the histograms of observations from the Sound Classification dataset. Which one of the following statements is true?

- A. Attribute  $x_1$  has an empirical mean of  $\approx -3.72$  and attribute  $x_6$  has an empirical variance of  $\approx 2.34$ .
- B. Attribute  $x_2$  has an empirical mean of  $\approx 4.76$  and attribute  $x_1$  has an empirical variance of  $\approx 0.12$ .
- C. Attribute  $x_3$  has an empirical mean of  $\approx 27.43$  and attribute  $x_1$  has an empirical variance of  $\approx 9.6$ .
- D. Attribute  $x_4$  has an empirical mean of  $\approx 8.73$  and attribute  $x_3$  has an empirical variance of  $\approx 0.27$ .
- E. Don't know.

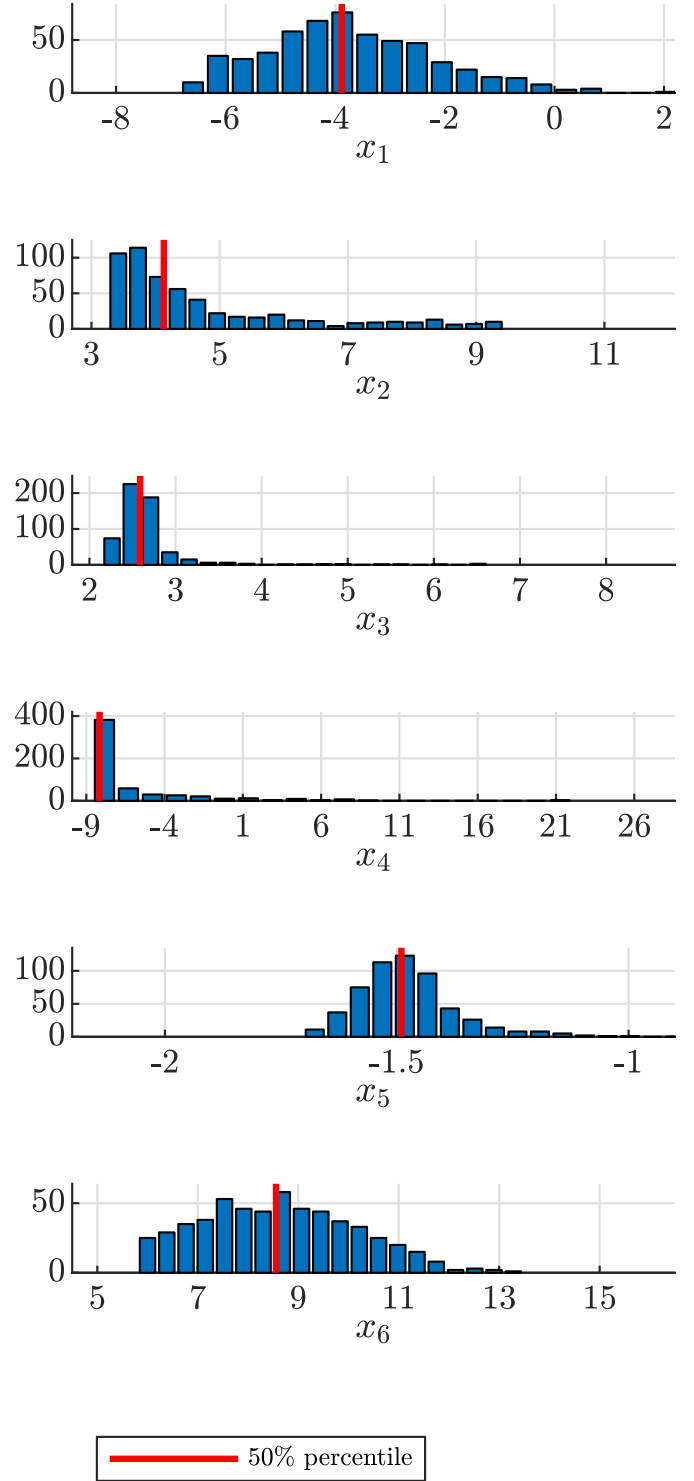


Figure 1: Histograms of observations from the Sound Classification dataset described in Table 1.

<sup>1</sup>Dataset obtained from <https://github.com/karolpiczak/ESC-50>

**Question 2.** The empirical correlation between the first 5 attributes in the Sound Classification dataset is given by the following correlation matrix

$$Corr = \begin{bmatrix} 1.0 & 0.48 & -0.14 & 0.15 & -0.06 \\ 0.48 & 1.0 & -0.14 & 0.19 & 0.19 \\ -0.14 & -0.14 & 1.0 & 0.12 & 0.0 \\ 0.15 & 0.19 & 0.12 & 1.0 & 0.26 \\ -0.06 & 0.19 & 0.0 & 0.26 & 1.0 \end{bmatrix},$$

where element  $Corr[i, j]$  is the correlation between the two attributes  $x_i$  and  $x_j$ , e.g.,  $Corr[x_2, x_5] = 0.19$ .

Additionally, the empirical standard deviation of  $x_1$  is 1.55, and the empirical standard deviation of  $x_4$  is 3.78.

Determine which one of the following statements is true.

- A. The covariance between  $x_1$  and  $x_4$  is  $\approx 0.88$
- B. The covariance between  $x_1$  and  $x_4$  is  $\approx 0.36$
- C. The covariance between  $x_1$  and  $x_4$  is  $\approx 1.36$
- D. The covariance between  $x_1$  and  $x_4$  is  $\approx 0.03$
- E. Don't know.

**Question 3.** A Principal Component Analysis (PCA) is carried out on the Sound Classification dataset in Table 1 based on the attributes  $x_1, x_2, x_3, x_4$  and  $x_5$ .

The data is standardized by (i) subtracting the mean and (ii) dividing each column by its standard deviation to obtain the standardized data matrix  $\tilde{X}$ . A singular value decomposition (SVD) is then carried out on the standardized data matrix to obtain the decomposition  $USV^T = \tilde{X}$

$$V = \begin{bmatrix} -0.5939 & 0.2906 & v_{1,3} & 0.0621 & 0.6652 \\ -0.6521 & 0.0759 & 0.0004 & 0.3813 & v_{2,5} \\ 0.2028 & -0.5105 & -0.7036 & 0.4508 & 0.0010 \\ -0.3696 & -0.5414 & -0.1781 & -0.7244 & -0.1173 \\ -0.2102 & -0.5967 & 0.5973 & 0.3503 & 0.3467 \end{bmatrix} \quad (1)$$

$$S = \begin{bmatrix} 30.3832 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & \sigma_{2,2} & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 22.7730 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 19.7263 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 16.0724 \end{bmatrix} \quad (2)$$

It is noted that certain elements are missing in the SVD, specifically  $v_{1,3}, v_{2,5}$  and  $\sigma_{2,2}$ . Additionally,  $\|\tilde{X}\|_F^2 = 2814.8909$ .

Determine which one of the following statements is correct.

- A.  $v_{1,3} \approx -0.1399, v_{2,5} \approx -0.2116$   
 $\sigma_{2,2} \approx 5.1903$
- B.  $v_{1,3} \approx -0.3413, v_{2,5} \approx -0.6508$   
 $\sigma_{2,2} \approx 26.9387$
- C.  $v_{1,3} \approx 0.3425, v_{2,5} \approx -0.6506$   
 $\sigma_{2,2} \approx 26.9387$
- D.  $v_{1,3} \approx -1.8385, v_{2,5} \approx -0.1629$   
 $\sigma_{2,2} \approx 5.1903$
- E. Don't know.

**Question 4.** Consider again the PCA for the Sound Classification dataset, in particular the SVD of  $\tilde{X}$  in Equation (1). Which one of the following statements is true?

- A. An observation with a low value of  $x_1$  (**P2AR**), a high value of  $x_2$  (**PER**), a low value of  $x_3$  (**RO**), a high value of  $x_4$  (**FLU**), and a high value of  $x_5$  (**SHP**) will typically have a positive value of the projection onto principal component number 1.
- B. An observation with a high value of  $x_2$  (**PER**), a high value of  $x_3$  (**RO**), a low value of  $x_4$  (**FLU**), and a high value of  $x_5$  (**SHP**) will typically have a negative value of the projection onto principal component number 4.
- C. An observation with a low value of  $x_1$  (**P2AR**), a high value of  $x_3$  (**RO**), a high value of  $x_4$  (**FLU**), and a high value of  $x_5$  (**SHP**) will typically have a positive value of the projection onto principal component number 2.
- D. An observation with a high value of  $x_1$  (**P2AR**), a high value of  $x_2$  (**PER**), a low value of  $x_3$  (**RO**), a high value of  $x_4$  (**FLU**), and a high value of  $x_5$  (**SHP**) will typically have a negative value of the projection onto principal component number 1.
- E. Don't know.

|          | $o_1$ | $o_2$ | $o_3$ | $o_4$ | $o_5$ | $o_6$ | $o_7$ | $o_8$ | $o_9$ | $o_{10}$ |
|----------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|
| $o_1$    | 0.0   | 1.3   | 4.1   | 3.8   | 4.5   | 2.4   | 3.2   | 2.7   | 3.0   | 3.9      |
| $o_2$    | 1.3   | 0.0   | 3.2   | 3.1   | 4.7   | 2.3   | 2.6   | 2.2   | 2.7   | 4.2      |
| $o_3$    | 4.1   | 3.2   | 0.0   | 0.4   | 4.9   | 2.7   | 1.1   | 1.6   | 2.4   | 4.8      |
| $o_4$    | 3.8   | 3.1   | 0.4   | 0.0   | 4.6   | 2.5   | 0.9   | 1.3   | 2.1   | 4.5      |
| $o_5$    | 4.5   | 4.7   | 4.9   | 4.6   | 0.0   | 3.1   | 4.4   | 3.7   | 2.8   | 2.3      |
| $o_6$    | 2.4   | 2.3   | 2.7   | 2.5   | 3.1   | 0.0   | 1.8   | 1.2   | 0.9   | 2.8      |
| $o_7$    | 3.2   | 2.6   | 1.1   | 0.9   | 4.4   | 1.8   | 0.0   | 1.0   | 1.7   | 4.1      |
| $o_8$    | 2.7   | 2.2   | 1.6   | 1.3   | 3.7   | 1.2   | 1.0   | 0.0   | 1.1   | 3.6      |
| $o_9$    | 3.0   | 2.7   | 2.4   | 2.1   | 2.8   | 0.9   | 1.7   | 1.1   | 0.0   | 2.9      |
| $o_{10}$ | 3.9   | 4.2   | 4.8   | 4.5   | 2.3   | 2.8   | 4.1   | 3.6   | 2.9   | 0.0      |

Table 2: The pairwise Euclidian distances,  $d(o_i, o_j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2 = \sqrt{\sum_{k=1}^M (x_{ik} - x_{jk})^2}$  between 10 observations from the Sound Classification dataset (recall that  $M = 6$ ). Each observation  $o_i$  corresponds to a row of the data matrix  $\mathbf{X}$  of Table 1. The colors indicate classes such that the black observations  $\{o_1, o_2, o_3, o_4\}$  belong to class  $C_1$  (corresponding to Machine), and the red observations  $\{o_5, o_6, o_7, o_8, o_9, o_{10}\}$  belong to class  $C_2$  (corresponding to Natural). To avoid single features to dominate, the dataset was standardized by subtracting the mean and dividing by the standard deviation.

**Question 5.** Consider the distances in Table 2 based on 10 observations from the Sound Classification dataset. To examine if observation  $o_1$  may be an outlier, we will calculate the  $K$ -nearest neighborhood density using only the observations and distances in Table 2. For an observation  $o_i$ , recall the density is computed using the set of  $K$  nearest neighbors of observation  $o_i$  excluding the  $i$ 'th observation itself,  $N_{\mathbf{X}_{\setminus i}}(o_i, K)$ , and is denoted by density  $\mathbf{X}_{\setminus i}(o_i, K)$ . What is the density for observation  $o_1$  for  $K = 3$  nearest neighbors?

- A. 0.732
- B. 0.769
- C. 0.469
- D. 0.640
- E. Don't know.

|         | $j = 1$ | $j = 2$ | $j = 3$ | $j = 4$ | $j = 5$ |
|---------|---------|---------|---------|---------|---------|
| $K = 1$ | 0       | 0       | 1       | 0       | 0       |
| $K = 3$ | 1       | 1       | 1       | 1       | 1       |
| $K = 4$ | 1       | 1       | 1       | 0       | 0       |

Table 3: Error rates corresponding to a specific inner fold,  $j$ , and model complexity,  $K$ , in **outer fold 1**.

|         | $j = 1$ | $j = 2$ | $j = 3$ | $j = 4$ | $j = 5$ |
|---------|---------|---------|---------|---------|---------|
| $K = 1$ | 0       | 0       | 1       | 0       | 1       |
| $K = 3$ | 0       | 0       | 1       | 0       | 0       |
| $K = 4$ | 0       | 0       | 1       | 0       | 0       |

Table 4: Error rates corresponding to a specific inner fold,  $j$ , and model complexity,  $K$ , in **outer fold 2**.

**Question 6.** We consider the generalization error of  $K$ -nearest neighbour (KNN) classifiers with  $K = [1, 3, 4]$  neighbors based on the observations in Table 2.

We apply two-layer cross-validation (cf. Algorithm 6 in the lecture notes) to estimate the generalization error with 2 outer and 5 inner folds. Table 3 and Table 4 provides partial results of the associated two-layer cross validation. It is noted that with other things being equal, a model with the lowest value of  $K$  is preferred in this context.

The specific split of the dataset in the two outer folds is also known:

- Outer fold  $i = 1$  training set:  $o_1, o_2, o_4, o_8, o_9$ .
- Outer fold  $i = 2$  training set:  $o_3, o_5, o_6, o_7, o_{10}$ .

Determine the error rates on the test set in each of the 2 outer folds based on the information in Table 3, Table 4 and Table 2.

- A.  $E_{i=1}^{test} = 0.1$  and  $E_{i=2}^{test} = 0.3$
- B.  $E_{i=1}^{test} = 0.2$  and  $E_{i=2}^{test} = 0.2$
- C.  $E_{i=1}^{test} = 0.8$  and  $E_{i=2}^{test} = 0.4$
- D.  $E_{i=1}^{test} = 0.2$  and  $E_{i=2}^{test} = 0.6$
- E. Don't know.

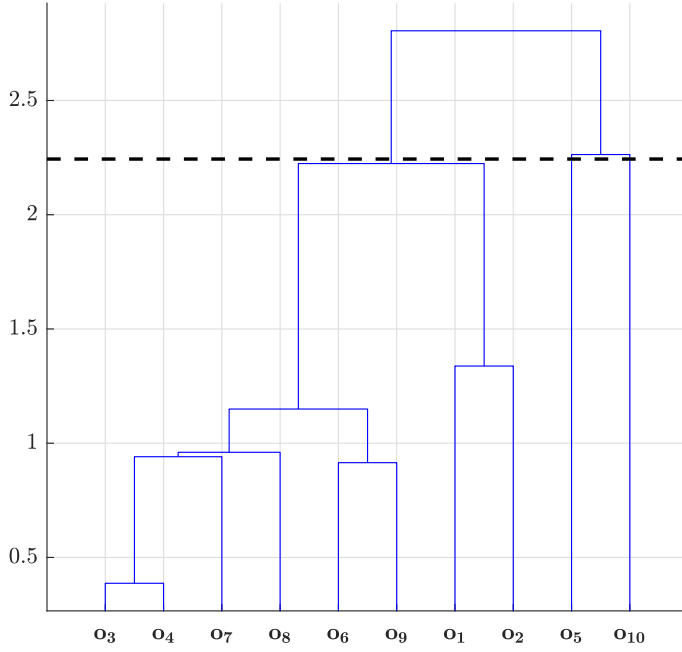


Figure 2: A dendrogram with a cutoff indicated by the dotted line, thereby generating 3 clusters.

**Question 7.** Consider the dendrogram shown in Figure 2 which is based on the observations and distances in Table 2 using single linkage.

Suppose we apply a cutoff (indicated by the black line in Figure 2) thereby generating three clusters. We wish to compare the quality of this clustering,  $Q$ , to the ground-truth clustering,  $Z$ , indicated by the colors in Table 2. What is the Rand index of the two clusterings?

- A.  $R(Z, Q) \approx 0.428$
- B.  $R(Z, Q) \approx 0.444$
- C.  $R(Z, Q) \approx 0.48$
- D.  $R(Z, Q) \approx 0.411$
- E. Don't know.

**Question 8.** Consider the ten observations in Table 2 and denote the partition implied by the ground truth classification by  $Z$ . We also consider an alternative partition  $Q$ , where all ten observations are placed in the same cluster. Let  $R(Z, Q)$  denote the Rand index and  $J(Z, Q)$  the Jaccard similarity between  $Z$  and  $Q$ . Which one of the following statements is true?

- A.  $R(Z, Q) > J(Z, Q)$
- B.  $R(Z, Q) = J(Z, Q)$
- C.  $R(Z, Q) < J(Z, Q)$
- D.  $R(Z, Q)$  and  $J(Z, Q)$  are not defined in this case
- E. Don't know.

**Question 9.** We consider a synthetic 1-dimensional dataset with the following  $N = 5$  observations:

$$\mathbf{X} = \begin{bmatrix} 2 \\ 5 \\ 8 \\ 12 \\ 13 \end{bmatrix}$$

Suppose we want to perform  $K$ -means clustering with  $K = 2$  centroids. The centroids are initialized at locations  $\mu_1 = 4$  and  $\mu_2 = 10$ . What will be the total cost (sum of squared distances),  $E$ , after the first iteration of  $K$ -means?

- A.  $E \approx 18.5$
- B.  $E \approx 9.0$
- C.  $E \approx 3.7$
- D.  $E \approx 299.75$
- E. Don't know.

| $p(\hat{x}_2, \hat{x}_4, y)$   | $y = \text{Machine}$ | $y = \text{Natural}$ |
|--------------------------------|----------------------|----------------------|
| $\hat{x}_2 = 0, \hat{x}_4 = 0$ | 0.18                 | 0.08                 |
| $\hat{x}_2 = 0, \hat{x}_4 = 1$ | 0.17                 | 0.07                 |
| $\hat{x}_2 = 1, \hat{x}_4 = 0$ | 0.08                 | 0.16                 |
| $\hat{x}_2 = 1, \hat{x}_4 = 1$ | 0.1                  | 0.16                 |

Table 5: Probability of observing particular values of  $\hat{x}_2$ ,  $\hat{x}_4$ , and  $y$ .

**Question 10.** Consider the Sound Classification dataset from Table 1. Suppose the attributes have been binarized such that  $\hat{x}_2 = 0$  corresponds to  $x_2 \leq -0.405$  (and otherwise  $\hat{x}_2 = 1$ ) and  $\hat{x}_4 = 0$  corresponds to  $x_4 \leq -0.456$  (and otherwise  $\hat{x}_4 = 1$ ). Suppose the probability for each of the configurations of  $\hat{x}_2$ ,  $\hat{x}_4$  and  $y$  are as given in Table 5.

What is then the probability  $y$  corresponds to *Machine* given an observation has  $\hat{x}_2 = 1$ ?

- A.  $p(y = \text{Machine} \mid \hat{x}_2 = 1) = 0.18$
- B.  $p(y = \text{Machine} \mid \hat{x}_2 = 1) = 0.34$
- C.  $p(y = \text{Machine} \mid \hat{x}_2 = 1) = 0.36$
- D.  $p(y = \text{Machine} \mid \hat{x}_2 = 1) = 0.261$
- E. Don't know.

**Question 11.** We wish to predict which of the two classes an observation  $\mathbf{x}$  belongs to in the Sound Classification dataset described in Table 1. To accomplish this we apply a Naïve-Bayes classifier where we model each of the  $M = 6$  features using a 1-dimensional normal distribution. The classifier will be used in an embedded setting where model prediction speed is paramount. Therefore, consider a single model evaluation:

$$p(y = \text{Machine} \mid \mathbf{x}).$$

What is the minimum number of evaluations of the normal density function  $\mathcal{N}(x \mid \mu, \sigma^2)$  we have to perform to compute this quantity?

- A. 14
- B. 21
- C. 12
- D. 18
- E. Don't know.

|               | $M_A$ correct | $M_A$ wrong |
|---------------|---------------|-------------|
| $M_B$ correct | 416           | 42          |
| $M_B$ wrong   | 38            | 68          |
| $M_C$ correct | 68            | 38          |
| $M_C$ wrong   | 42            | 416         |

Table 6: Outcome of cross-validation with paired test observations. The values indicate the number of times where two models both classify correctly, both classify incorrectly (i.e., both wrong), or one classifies correctly and the other one incorrectly.

**Question 12.** Consider the comparison of a classification model,  $M_A$ , with two alternative models,  $M_B$  and  $M_C$ , using McNemar's test as defined in Section 11.3.3 of the lecture notes.

The test is based on the output of three-fold cross-validation on the Sound Classification dataset with the relevant results of the cross-validation shown in Table 6.

The comparison between  $M_A$  and  $M_B$  results in a p-value ( $p_{AB}$ ) and the difference in accuracy ( $\hat{\theta}_{AB}$ ).

Similarly, the comparison between  $M_A$  and  $M_C$  results in a p-value ( $p_{AC}$ ) and the difference in accuracy ( $\hat{\theta}_{AC}$ ). Determine which one of the following statements is correct.

*Hint: The problem can be solved without evaluating complicated mathematical functions.*

- A.  $p_{AB} = p_{AC}$  and  $\hat{\theta}_{AB} = -\hat{\theta}_{AC}$ .
- B.  $p_{AB} \neq p_{AC}$  and  $\hat{\theta}_{AB} = \hat{\theta}_{AC}$ .
- C.  $p_{AB} = p_{AC}$  and  $\hat{\theta}_{AB} = \hat{\theta}_{AC}$ .
- D.  $p_{AB} \neq p_{AC}$  and  $\hat{\theta}_{AB} = -\hat{\theta}_{AC}$ .
- E. Don't know.

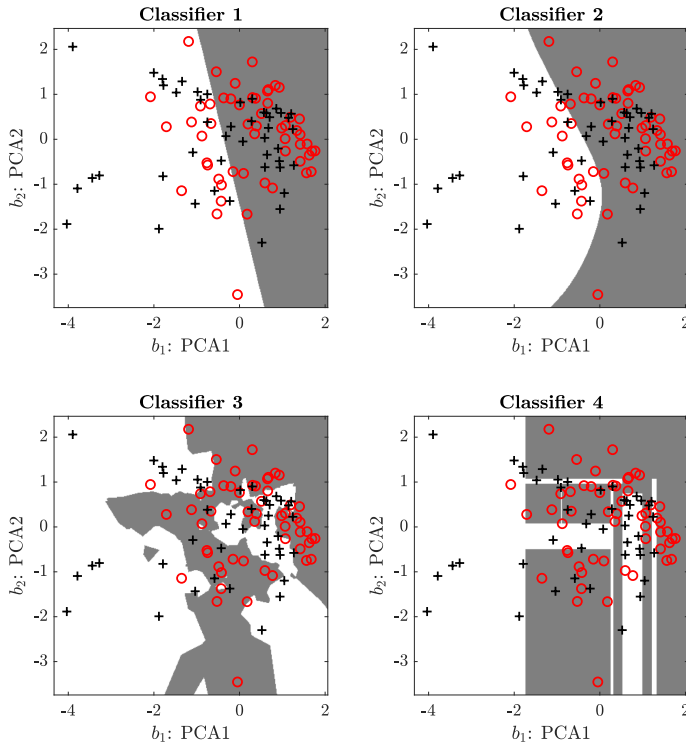


Figure 3: Decision boundaries for four different classifiers trained on the Sound Classification dataset when projected onto the first two principal components.

**Question 13.** Consider a subset of the Sound Classification dataset (described in Table 1) after it has been projected onto the first two principal components, thereby giving rise to a two-dimensional dataset with each observation having two coordinates,  $b_1$  and  $b_2$ .

We will consider the following four classifiers:

**LR:** Logistic regression

**ANN:** Artificial neural network with 5 hidden units

**CT:** Classification tree with regular axis-aligned splits

**KNN:** K-nearest neighbours with  $K = 3$

Suppose the classifiers are trained on the two-dimensional dataset and the decision boundary for each of the four classifiers is given in Figure 3. Which one of the following statements is correct?

- A. Classifier 1 corresponds to **LR**,  
Classifier 2 corresponds to **ANN**,  
Classifier 3 corresponds to **KNN**,  
Classifier 4 corresponds to **CT**.

- B. Classifier 1 corresponds to **ANN**,  
Classifier 2 corresponds to **KNN**,  
Classifier 3 corresponds to **LR**,  
Classifier 4 corresponds to **CT**.
- C. Classifier 1 corresponds to **KNN**,  
Classifier 2 corresponds to **CT**,  
Classifier 3 corresponds to **ANN**,  
Classifier 4 corresponds to **LR**.
- D. Classifier 1 corresponds to **LR**,  
Classifier 2 corresponds to **KNN**,  
Classifier 3 corresponds to **ANN**,  
Classifier 4 corresponds to **CT**.
- E. Don't know.

**Question 14.** Determine which one of the following statements is always true about two-layer cross-validation with  $K_1$  outer folds,  $K_2$  inner folds,  $S$  model complexities, and a dataset of size  $N$  (cf. Algorithm 6 in the lecture notes).

- A. The total number of times each observation is used for testing is  $K_1 \times (K_2 \times S + 1)$ .
- B. The total number of times each observation is used for training is  $(K_1 - 1) \times ((K_2 - 1) \times S + 1)$ .
- C. The total number of models needed to be trained is  $K_2 \times S \times (K_1 + 1)$ .
- D. The size of the test set in each of the  $K_1$  outer folds is always greater than the test set in the  $K_2$  inner folds.
- E. Don't know.

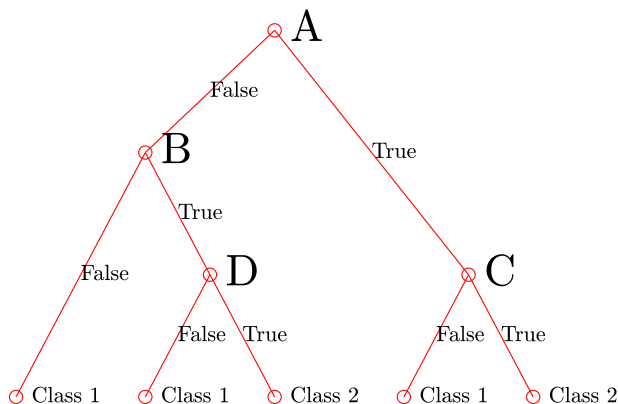


Figure 4: Classification tree.

**Question 15.** Consider an artificial dataset of  $N = 4000$  observations. The dataset is classified according to a decision tree of the form shown in Figure 4 resulting in a partition into classes indicated by the colors/markers in Figure 5. What is the correct rule assignment to the nodes in the decision tree?

A.  $A: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right\|_1 < 3, B: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 6 \\ 2 \end{bmatrix} \right\|_2 < 3,$   
 $C: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right\|_2 < 2, D: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right\|_1 < 3$

B.  $A: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 6 \\ 2 \end{bmatrix} \right\|_2 < 3, B: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right\|_1 < 3,$   
 $C: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right\|_1 < 3, D: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right\|_2 < 2$

C.  $A: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right\|_1 < 3, B: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right\|_1 < 3,$   
 $C: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 6 \\ 2 \end{bmatrix} \right\|_2 < 3, D: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right\|_2 < 2$

D.  $A: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right\|_2 < 2, B: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right\|_1 < 3,$   
 $C: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 6 \\ 2 \end{bmatrix} \right\|_2 < 3, D: \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right\|_1 < 3$

E. Don't know.

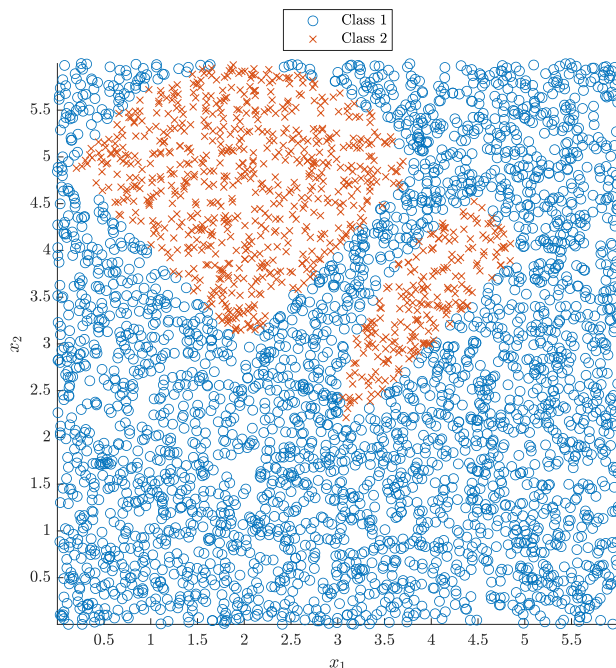


Figure 5: Classification boundary.

**Question 16.** Consider standard linear regression (no regularization) with the prediction rule  $f(\mathbf{x}_i; \mathbf{w}) = \tilde{\mathbf{x}}_i^\top \mathbf{w}$ . Here  $\tilde{\mathbf{x}}_i^\top = [1 \ \mathbf{x}_i^\top]$  is the  $i$ 'th observation,  $\mathbf{x}_i$ , concatenated with a constant with a value of 1. Suppose, the weights,  $\mathbf{w} = [w_0 \ w_1 \ \dots \ w_M]^\top$  (including the intercept term  $w_0$ ), were learned by predicting  $y_r$  from  $\tilde{\mathbf{X}} = [\tilde{\mathbf{x}}_1 \ \tilde{\mathbf{x}}_2 \ \dots \ \tilde{\mathbf{x}}_N]^\top$  ( $N$  is the number of observations) and minimizing the sum-of-squares error function, as usual.

What solution,  $\mathbf{w}_{\text{scaled}}$ , do you obtain if you scale all elements in  $\tilde{\mathbf{X}}$  by a factor of  $1/\beta$ , with  $\beta \in \mathbb{R}$  being a constant, before learning the weights?

A.  $\mathbf{w}_{\text{scaled}} = \frac{1}{\beta} \mathbf{w}$

B.  $\mathbf{w}_{\text{scaled}} = \mathbf{w}^\beta$

C.  $\mathbf{w}_{\text{scaled}} = \beta \mathbf{w}$

D.  $\mathbf{w}_{\text{scaled}} = \beta^2 \mathbf{w}$

E. Don't know.



**Question 17.** Suppose a neural network is trained to classify sounds. As part of training the network, we wish to select between eight different ways to transform the attributes (i.e.,  $S = 8$  models) and estimate the generalization error of the optimal choice. In the outer loop we opt for  $K_1 = 3$ -fold cross-validation, and in the inner  $K_2 = 5$ -fold cross-validation. The time taken to *train* a single model is 20 minutes, and this can be assumed constant for each fold. If the time taken to test a model is negligible, what is the total time required for the 2-level cross-validation procedure?

- A. 360 minutes
- B. 2460 minutes
- C. 4920 minutes
- D. 2400 minutes
- E. Don't know.

**Question 18.** Suppose a logistic regression model has been trained on a two-dimensional dataset based on two attributes from the the Sound Classification dataset. Now consider four test observations:

$$\begin{aligned}\mathbf{x}_1 &= [1 \quad 1]^\top \\ \mathbf{x}_2 &= [-1.0 \quad 0.6]^\top \\ \mathbf{x}_3 &= [-0.5 \quad -0.5]^\top \\ \mathbf{x}_4 &= [-0.91 \quad -0.16]^\top\end{aligned}$$

Additionally, the following probabilities are known

$$\begin{aligned}p(y = \text{Machine} \mid \mathbf{x}_1) &= 0.73 \\ p(y = \text{Machine} \mid \mathbf{x}_2) &= 0.47 \\ p(y = \text{Machine} \mid \mathbf{x}_3) &= 0.73\end{aligned}$$

Determine which one of the following statements is true.

*Hint: The problem can be solved without complicated numerical computation.*

- A.  $p(y = \text{Machine} \mid \mathbf{x}_4) < p(y = \text{Machine} \mid \mathbf{x}_2)$
- B.  $p(y = \text{Machine} \mid \mathbf{x}_4) = p(y = \text{Machine} \mid \mathbf{x}_1)$
- C.  $p(y = \text{Machine} \mid \mathbf{x}_4) > p(y = \text{Machine} \mid \mathbf{x}_2)$
- D.  $p(y = \text{Machine} \mid \mathbf{x}_4) > p(y = \text{Machine} \mid \mathbf{x}_1)$
- E. Don't know.

|           | $i = 1$ | $i = 2$ | $i = 3$ | $i = 4$ |
|-----------|---------|---------|---------|---------|
| $x_i$     | -0.5    | 0.39    | 1.19    | -1.08   |
| $y_{r,i}$ | -0.86   | -0.61   | 1.37    | 0.1     |

Table 7: A subset of the Sound Classification dataset.

**Question 19.** Consider once again the Sound Classification dataset, but this time we limit ourselves to  $N = 4$  observations and a single attribute from the full dataset with the  $i$ 'th observation denoted by  $x_i$ . In particular, we consider transformations of the  $i$ 'th observation,  $x_i$ , resulting in  $\tilde{\mathbf{x}}_i$  (a column vector).

The goal is to predict  $y_r$ , and to achieve this; we will apply ridge regression with the prediction rule  $f(\tilde{\mathbf{x}}_i; w_0, \mathbf{w}) = w_0 + \tilde{\mathbf{x}}_i^\top \mathbf{w}$ . Ridge regression determines the constant offset,  $w_0$ , and weights  $\mathbf{w} = [w_1 \quad w_2 \quad \dots \quad w_M]^\top$  ( $M$  is the dimensionality of  $\tilde{\mathbf{x}}_i$ ), by minimizing a cost function of the form:

$$E_\lambda = \sum_{i=1}^N (y_{r,i} - f(\tilde{\mathbf{x}}_i; w_0, \mathbf{w}))^2 + \lambda \|\mathbf{w}\|_2^2$$

Suppose  $w_0$  and  $\mathbf{w}$  were learned based on the dataset in Table 7 using a specific transformation of  $x$ . The weights were learned with  $\lambda = 0.25$  resulting in  $E_{\lambda=0.25} \approx 0.2$ ,  $w_0 = 0.0$  and  $\mathbf{w} = [0.39 \quad 0.77]^\top$ .

Standardization was performed on the data matrix with the transformed observations,  $\tilde{\mathbf{X}} = [\tilde{\mathbf{x}}_1 \quad \tilde{\mathbf{x}}_2 \quad \tilde{\mathbf{x}}_3 \quad \tilde{\mathbf{x}}_4]^\top$ , prior to learning the parameters by subtracting the column-wise mean and dividing by the unbiased estimate of the column-wise standard deviation.

Which one of the following transformations was used when learning  $w_0$  and  $\mathbf{w}$ ?

- A.  $\tilde{\mathbf{x}}_i = [x_i \quad x_i^3]^\top$
- B.  $\tilde{\mathbf{x}}_i = [x_i \quad \sin(x_i) \quad x_i^2]^\top$
- C.  $\tilde{\mathbf{x}}_i = [x_i \quad \sin(x_i)]^\top$
- D.  $\tilde{\mathbf{x}}_i = [x_i \quad x_i^2]^\top$
- E. Don't know.

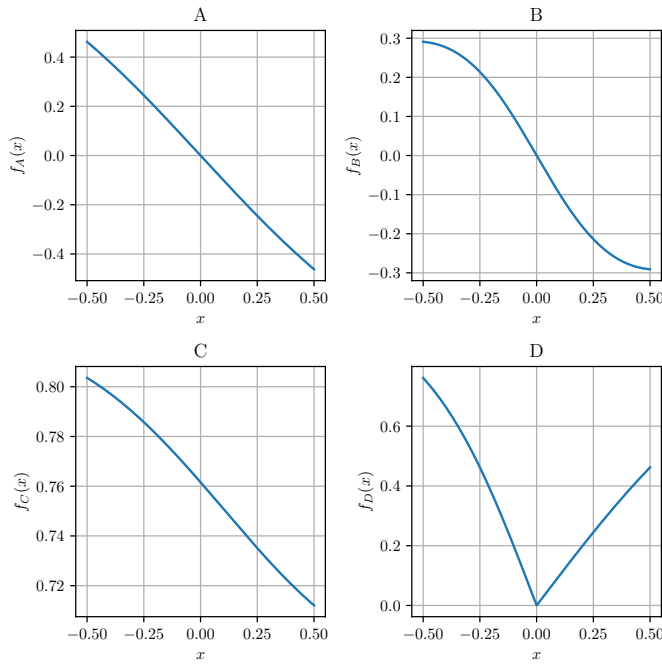


Figure 6: The output of a neural network  $f(x)$  as a function of the input  $x$  for four different choices of the activation function  $h^{(1)}$  for the hidden layer.

**Question 20.** Consider a two-layer neural network  $f : \mathbb{R} \rightarrow \mathbb{R}$  for regression with two hidden units (i.e.  $\mathbf{z}^{(1)} \in \mathbb{R}^2$ ) and of the form

$$\begin{aligned} \mathbf{z}^{(1)} &= h^{(1)}(\tilde{\mathbf{x}}\mathbf{W}^{(1)}), \\ f(\mathbf{x}) &= h^{(2)}(\tilde{\mathbf{z}}^{(1)}\mathbf{W}^{(2)}), \end{aligned}$$

where  $\tilde{\mathbf{x}} = [1 \ x]$ ,  $\tilde{\mathbf{z}}^{(1)} = [1 \ z_1^{(1)} \ z_2^{(1)}]$ ,  $h^{(1)}(x)$  is the activation function for the hidden layer that is applied elementwise, and  $h^{(2)}(x)$  is the activation function of the output layer.

Assume that  $h^{(2)}(x)$  is the hyperbolic tangent activation function and the network's weights are given by

$$\mathbf{W}^{(1)} = \begin{bmatrix} 0 & 0 \\ 1 & -2 \end{bmatrix} \quad \text{and} \quad \mathbf{W}^{(2)} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

Figure 6 shows  $f(x)$  for four different choices of the activation function  $h^{(1)}$ . Which one of the following statements is correct?

- A. For  $f_A$ ,  $h^{(1)}$  is the **rectified linear unit**  
For  $f_B$ ,  $h^{(1)}$  is the **identity function**  
For  $f_C$ ,  $h^{(1)}$  is the **hyperbolic tangent**  
For  $f_D$ ,  $h^{(1)}$  is the **logistic sigmoid**
- B. For  $f_A$ ,  $h^{(1)}$  is the **identity function**  
For  $f_B$ ,  $h^{(1)}$  is the **logistic sigmoid**  
For  $f_C$ ,  $h^{(1)}$  is the **hyperbolic tangent**  
For  $f_D$ ,  $h^{(1)}$  is the **rectified linear unit**
- C. For  $f_A$ ,  $h^{(1)}$  is the **identity function**  
For  $f_B$ ,  $h^{(1)}$  is the **hyperbolic tangent**  
For  $f_C$ ,  $h^{(1)}$  is the **logistic sigmoid**  
For  $f_D$ ,  $h^{(1)}$  is the **rectified linear unit**
- D. For  $f_A$ ,  $h^{(1)}$  is the **hyperbolic tangent**  
For  $f_B$ ,  $h^{(1)}$  is the **identity function**  
For  $f_C$ ,  $h^{(1)}$  is the **logistic sigmoid**  
For  $f_D$ ,  $h^{(1)}$  is the **rectified linear unit**
- E. Don't know.

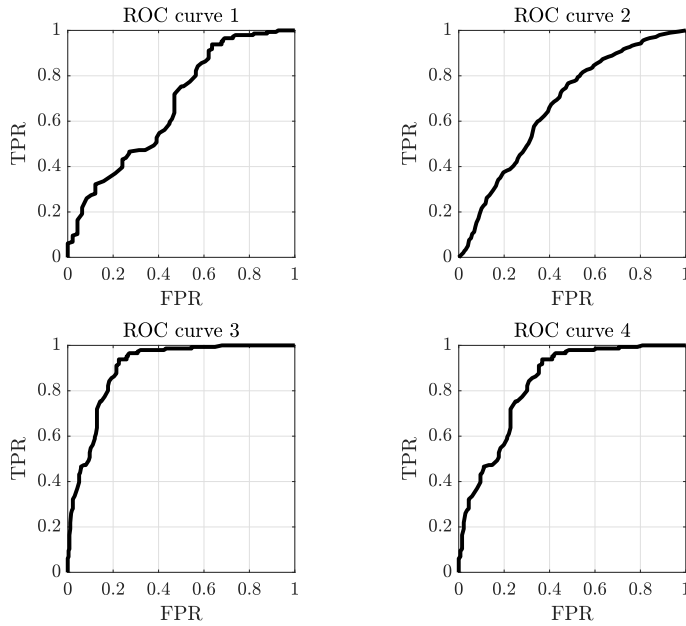


Figure 7: Candidate ROC curves for the classifier.

**Question 21.** We wish to predict whether an observation from the Sound Classification dataset (see Table 1) belongs to the Machine class (or not). To accomplish this, we fit a logistic regression model to the dataset, and for each observation  $\mathbf{x}_i, y_i$  obtain a class-probability prediction  $\hat{y}_i \in [0, 1]$ . We threshold the class-probability at different values  $\theta$  thereby obtaining, for each value of  $\theta$ , the true positives (TP), true negatives (TN), false positives (FP) and false negatives (FN). These are plotted as functions of  $\theta$  in Figure 8. Which of the receiver-operator characteristic (ROC) plots shown in Figure 7 corresponds to these graphs?

- A. ROC curve 1
- B. ROC curve 2
- C. ROC curve 3
- D. ROC curve 4
- E. Don't know.

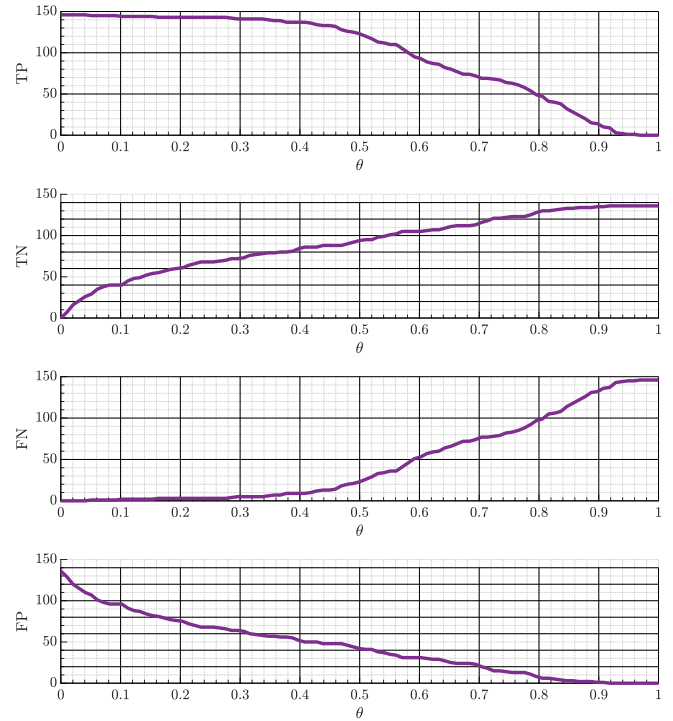


Figure 8: TP, TN, FN, and FP as functions of the threshold value  $\theta$ .

**Question 22.** In a market basket problem, we consider two item sets  $X$  and  $Y$ . Assume that the support of  $X$  is  $\frac{3}{5}$ , the support of  $Y$  is  $\frac{8}{15}$ , and the confidence of the association rule  $X \rightarrow Y$  is  $\frac{1}{6}$ . What is the confidence of  $Y \rightarrow X$ ?

- A.  $\text{conf}(Y \rightarrow X) = \frac{1}{10}$
- B.  $\text{conf}(Y \rightarrow X) = \frac{4}{27}$
- C.  $\text{conf}(Y \rightarrow X) = \frac{1}{6}$
- D.  $\text{conf}(Y \rightarrow X) = \frac{3}{16}$
- E. Don't know.

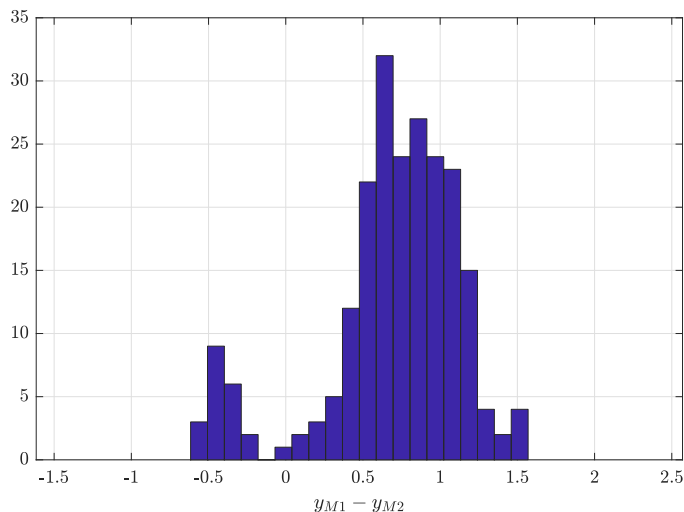


Figure 9: Histogram of  $y_{M1} - y_{M2}$  based on all test observations.

**Question 23.** Consider a statistical procedure to determine if there is a difference between two regression models, M1 and M2, using a paired test as described in Section 11.3.6 of the lecture notes.

The distribution of the paired differences in predictions between the two models,  $y_{M1} - y_{M2}$ , using a 5-fold cross validation setup is illustrated as a histogram in Figure 9.

Determine which combination of estimated difference ( $\hat{z}$ ), confidence interval (CI) and p-value is the only plausible answer.

- A.  $\hat{z} = 0.69$ ,  $CI = [0.63, 0.75]$ , p-value < 0.05
- B.  $\hat{z} = -1.05$ ,  $CI = [-1.29, -0.81]$ , p-value < 0.05
- C.  $\hat{z} = 0.63$ ,  $CI = [0.57, 0.69]$ , p-value < -0.05
- D.  $\hat{z} = 0.76$ ,  $CI = [0.59, 0.76]$ , p-value < 0.05
- E. Don't know.

|          | $b_1$ | $b_2$ | $b_3$ | $b_4$ | $b_5$ | $b_6$ |
|----------|-------|-------|-------|-------|-------|-------|
| $o_1$    | 1     | 1     | 0     | 1     | 0     | 0     |
| $o_2$    | 1     | 0     | 1     | 1     | 0     | 0     |
| $o_3$    | 0     | 0     | 1     | 1     | 1     | 1     |
| $o_4$    | 0     | 1     | 1     | 1     | 1     | 1     |
| $o_5$    | 1     | 1     | 0     | 1     | 1     | 0     |
| $o_6$    | 1     | 1     | 0     | 1     | 1     | 1     |
| $o_7$    | 0     | 1     | 1     | 1     | 0     | 1     |
| $o_8$    | 0     | 1     | 0     | 1     | 1     | 1     |
| $o_9$    | 0     | 1     | 0     | 1     | 1     | 1     |
| $o_{10}$ | 1     | 1     | 1     | 1     | 1     | 0     |

Table 8: Binarized version of the Sound Classification dataset. Each of the features  $b_i$  are obtained by taking a feature  $x_i$  and letting  $b_i = 1$  correspond to a value  $x_i$  greater than the median (otherwise  $b_i = 0$ ). The colors indicate classes such that the black observations  $\{o_1, o_2, o_3, o_4\}$  belong to class  $C_1$  (corresponding to Machine), and the red observations  $\{o_5, o_6, o_7, o_8, o_9, o_{10}\}$  belong to class  $C_2$  (corresponding to Natural).

**Question 24.** Consider again the Sound Classification dataset. Suppose we wish to predict the class label  $y$  using a decision tree model, and to improve performance, we wish to apply AdaBoost. We apply AdaBoost (as described in Algorithm 7 of the lecture notes) to the binarized version of the dataset in Table 8. Suppose that the classifier learned on dataset  $D_1$  in the first round of boosting is

$$f_1(b_1, b_2, b_3, b_4, b_5, b_6) = \begin{cases} C_1, & \text{if } b_3 = 1 \text{ and } b_4 = 1 \\ C_2, & \text{otherwise.} \end{cases}$$

What is the importance of the classifier  $f_1$ ?

- A.  $\alpha_1 \approx 0.84$
- B.  $\alpha_1 \approx 0.42$
- C.  $\alpha_1 \approx 0.70$
- D.  $\alpha_1 \approx 0.30$
- E. Don't know.

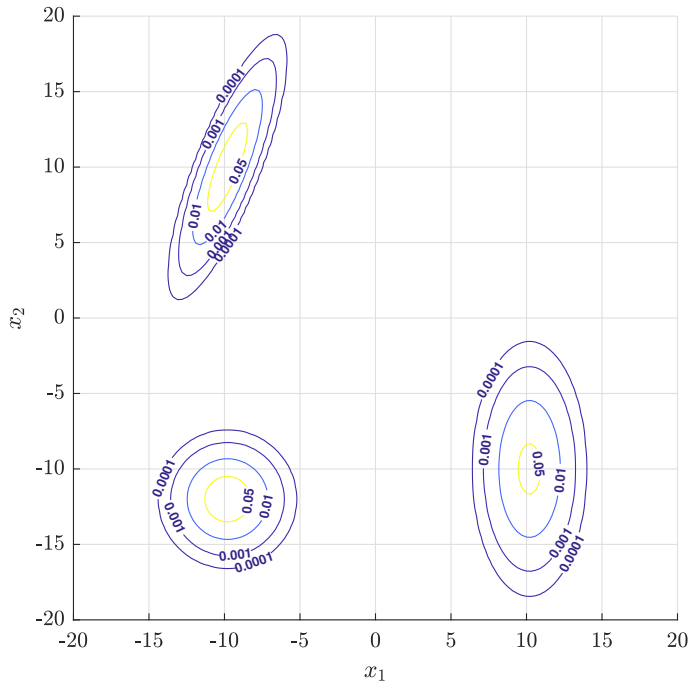


Figure 10: Contour lines corresponding to the density of the three individual components in the Gaussian Mixture Model.

**Question 25.** A Gaussian Mixture Model with  $K = 3$  components has been trained on a centered dataset (i.e., the column-wise mean has been subtracted) with two attributes. The model is defined as

$$p(\mathbf{x}) = \sum_{k=1}^3 w_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k).$$

The contour lines of the densities corresponding to the individual mixture components are shown in Figure 10.

The component weights and covariance matrices are:

$$w_1 = 0.5, \boldsymbol{\Sigma}_1 = \begin{bmatrix} 1.1 & 2.0 \\ 2.0 & 5.5 \end{bmatrix}$$

$$w_2 = 0.49, \boldsymbol{\Sigma}_2 = \begin{bmatrix} 1.1 & 0.0 \\ 0.0 & 5.5 \end{bmatrix}$$

$$w_3 = 0.01, \boldsymbol{\Sigma}_3 = \begin{bmatrix} 1.5 & 0.0 \\ 0.0 & 1.5 \end{bmatrix}$$

Determine which one of the following options corresponds to the first principal component direction,  $\mathbf{v}_1$ ,

and the second principal component direction,  $\mathbf{v}_2$ , of the dataset modeled by  $p(\mathbf{x})$ .

- A.  $\mathbf{v}_1 \approx [-0.3 \ -1.0]^\top$ ,  $\mathbf{v}_2 \approx [-1.0 \ 0.3]^\top$
- B.  $\mathbf{v}_1 \approx [-0.3 \ -1.0]^\top$ ,  $\mathbf{v}_2 \approx [0.0 \ 1.0]^\top$
- C.  $\mathbf{v}_1 \approx [0.7 \ -0.7]^\top$ ,  $\mathbf{v}_2 \approx [0.7 \ 0.7]^\top$
- D.  $\mathbf{v}_1 \approx [0.7 \ 0.7]^\top$ ,  $\mathbf{v}_2 \approx [0.7 \ -0.7]^\top$
- E. Don't know.

**Question 26.** Consider again the Sound Classification dataset in Table 1. We apply backward selection to find an interpretable linear regression model which uses a subset of the  $M = 6$  attributes to predict the Perceived annoyance measure (PAM)  $y_r$ . Recall backward selection chooses models based on the test error as determined by cross-validation, and in our case we use the hold-out method to generate a single test/training split.

Suppose backward selection ends up selecting the attributes  $x_1$ ,  $x_4$ ,  $x_5$ , and  $x_6$ , what is the minimal number of models which were *tested* in order to obtain this result?

- A. 15 models
- B. 19 models
- C. 16 models
- D. 10 models
- E. Don't know.

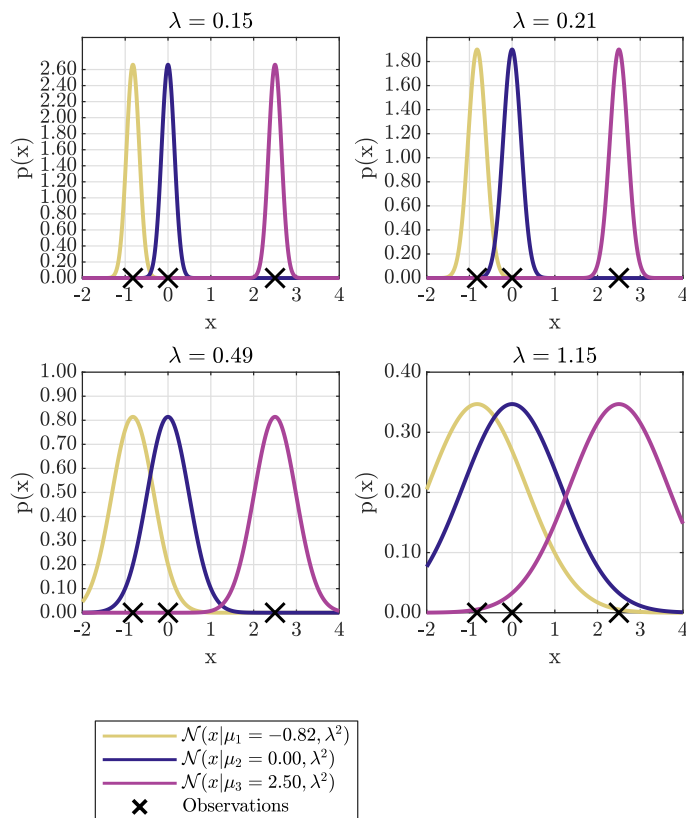


Figure 11: KDE applied to a 1-dimensional dataset using a Gaussian kernel with kernel widths  $\lambda = [0.15 \ 0.21 \ 0.49 \ 1.15]$ .

**Question 27.** A small 1-dimensional dataset with  $N = 3$  observations has been subsampled from the Sound Classification dataset, specifically:

$$\mathbf{X} = \begin{bmatrix} -0.82 \\ 0.0 \\ 2.5 \end{bmatrix}$$

A kernel density estimator (KDE) has been fitted to the small dataset with different  $\lambda$  using the standard Gaussian kernel (i.e., the individual Gaussian components in the KDE have variance  $\sigma^2 = \lambda^2$ ). The resulting densities are illustrated in Figure 11.

Consider a leave-one-out (LOO) procedure for computing the generalization error for different  $\lambda$  as in Section 20.1.1 of the lecture notes. The test log likelihood for the whole dataset is defined as

$$\begin{aligned} \mathcal{L}(\lambda) &= \frac{1}{N} \sum_{i=1}^N \log \left[ \sum_{j \neq i} \frac{1}{N-1} \mathcal{N}(x_i | x_j, \lambda^2) \right] \\ &= \frac{1}{N} \sum_{i=1}^N \log \left[ \sum_{j \neq i} \frac{1}{N-1} M_{ij} \right] = \frac{1}{N} \sum_{i=1}^N L_i \end{aligned}$$

i.e.  $L_i$  corresponds to the log density,  $\log p(x)$ , of the KDE constructed without  $x_i$  and evaluated at  $x_i$ .

Determine which value of  $\lambda$  in Figure 11 results in  $L = [-2.3 \ -2.3 \ -13.91]$ .

- A.  $\lambda = 1.15$
- B.  $\lambda = 0.15$
- C.  $\lambda = 0.21$
- D.  $\lambda = 0.49$
- E. Don't know.